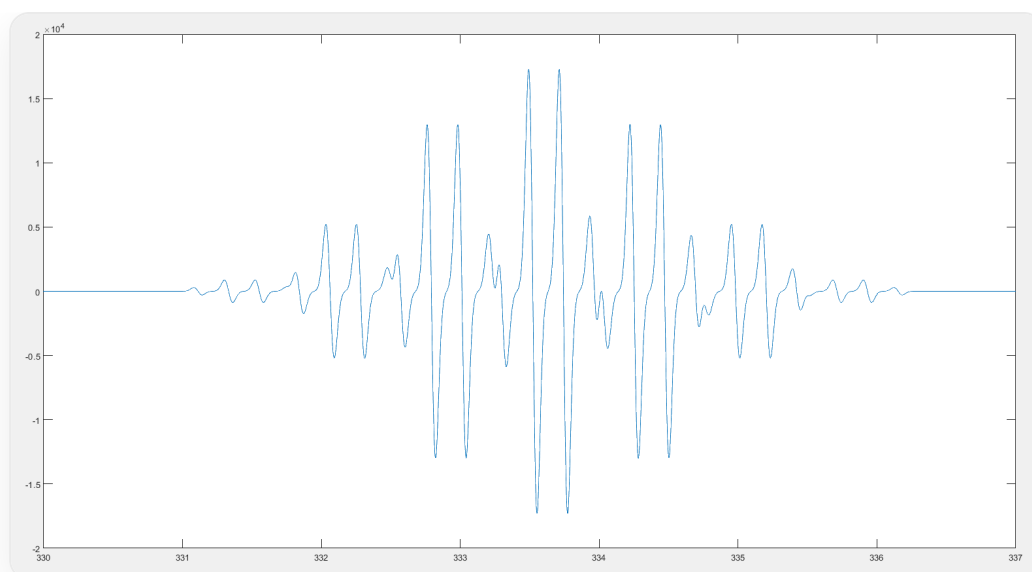


Question

There is an electrically neutral compound **X** with the chemical formula C_xH_y . All atoms of **X** are in the same plane. **X** is EPR active, and all protons contribute to hyperfine splitting. The simulated room-temperature continuous-wave EPR spectrum of its carbon tetrachloride solution is shown below. The microwave frequency was set to 9.35 GHz in the simulation.



This is an EPR spectrum with magnetic field strength on the horizontal axis and peak intensity on the vertical axis. The data points are as follows, with each pair of parentheses containing a data point. The first number is the magnetic field strength in mT, and the second number is the peak intensity (in arbitrary units):

(3.3000000e+02 -7.7775311e-13)(3.3000684e+02 -1.1674578e-12)(3.3001369e+02 -1.0367878e-12)
 (3.3002053e+02 -1.3620191e-12)(3.3002737e+02 -5.8390793e-13)(3.3003421e+02 -1.4261316e-12)
 (3.3004106e+02 2.5892558e-13)(3.3004790e+02 1.3036992e-13)(3.3005474e+02 -2.2054715e-12)
 (3.3006158e+02 1.2971828e-12)(3.3006843e+02 -1.2322962e-12)(3.3007527e+02 9.7259705e-13)
 (3.3008211e+02 7.7805246e-13)(3.3008895e+02 1.9453787e-12)(3.3009580e+02 1.2993699e-13)
 (3.3010264e+02 7.7808883e-13)(3.3010948e+02 2.5878850e-13)(3.3011632e+02 -3.9166737e-17)
 (3.3012317e+02 1.0382202e-12)(3.3013001e+02 4.5338344e-13)(3.3013685e+02 9.0795308e-13)
 (3.3014370e+02 9.7373568e-13)(3.3015054e+02 1.8154585e-12)(3.3015738e+02 1.1999126e-12)
 (3.3016422e+02 1.2007967e-12)(3.3017107e+02 -6.0069489e-13)(3.3017791e+02 7.8933339e-13)
 (3.3018475e+02 1.0719763e-13)(3.3019159e+02 6.7987822e-14)(3.3019844e+02 1.0378593e-12)
 (3.3020528e+02 1.4345950e-12)(3.3021212e+02 -3.0870176e-13)(3.3021896e+02 -2.0114187e-12)
 (3.3022581e+02 -1.0703617e-12)(3.3023265e+02 6.5894507e-14)(3.3023949e+02 -9.0831117e-13)
 (3.3024633e+02 3.8936894e-13)(3.3025318e+02 -1.8156152e-12)(3.3026002e+02 2.5838564e-13)
 (3.3026686e+02 -7.7851407e-13)(3.3027370e+02 -7.7852246e-13)(3.3028055e+02 -2.5990755e-13)
 (3.3028739e+02 1.8154362e-12)(3.3029423e+02 2.8535053e-12)(3.3030108e+02 -3.9014108e-13)
 (3.3030792e+02 7.2044232e-13)(3.3031476e+02 2.4641531e-12)(3.3032160e+02 7.7855883e-13)
 (3.3032845e+02 2.5869897e-13)(3.3033529e+02 -1.0376607e-12)(3.3034213e+02 1.0394008e-12)
 (3.3034897e+02 -7.7703133e-13)(3.3035582e+02 -1.8155369e-12)(3.3036266e+02 -1.0360492e-12)
 (3.3036950e+02 -1.4271863e-12)(3.3037634e+02 1.4922191e-12)(3.3038319e+02 -1.8805305e-12)
 (3.3039003e+02 -6.4886244e-13)(3.3039687e+02 3.2381015e-13)(3.3040371e+02 3.8925703e-13)
 (3.3041056e+02 -6.0791897e-13)(3.3041740e+02 1.2277542e-12)(3.3042424e+02 -4.8536831e-14)
 (3.3043109e+02 -4.7779733e-13)(3.3043793e+02 1.5082818e-12)(3.3044477e+02 -1.2001476e-12)

(3.3045161e+02 -4.5293582e-13)(3.3045846e+02 1.2990901e-13)(3.3046530e+02 1.1678159e-12)
 (3.3047214e+02 -7.7786502e-13)(3.3047898e+02 -5.2017319e-13)(3.3048583e+02 7.7804407e-13)
 (3.3049267e+02 2.5925850e-13)(3.3049951e+02 -6.7982265e-16)(3.3050635e+02 7.7758526e-13)
 (3.3051320e+02 -1.0386007e-12)(3.3052004e+02 -6.4903029e-13)(3.3052688e+02 1.9118508e-13)
 (3.3053372e+02 -2.6044749e-13)(3.3054057e+02 -1.0378061e-12)(3.3054741e+02 -1.2979543e-12)
 (3.3055425e+02 -2.5928647e-13)(3.3056109e+02 -1.5555398e-12)(3.3056794e+02 -5.1759938e-13)
 (3.3057478e+02 2.0760375e-12)(3.3058162e+02 2.6017612e-13)(3.3058847e+02 2.0754108e-12)
 (3.3059531e+02 1.1673682e-12)(3.3060215e+02 3.1130939e-12)(3.3060899e+02 8.7499463e-13)
 (3.3061584e+02 2.8532647e-12)(3.3062268e+02 1.8716447e-12)(3.3062952e+02 9.6984139e-14)
 (3.3063636e+02 1.5888205e-12)(3.3064321e+02 1.6026930e-13)(3.3065005e+02 -5.4155291e-13)
 (3.3065689e+02 -2.2534874e-12)(3.3066373e+02 -8.9077445e-14)(3.3067058e+02 9.8302789e-15)
 (3.3067742e+02 4.3817434e-13)(3.3068426e+02 -9.4035200e-13)(3.3069110e+02 7.1333575e-13)
 (3.3069795e+02 -2.6596433e-12)(3.3070479e+02 1.7502914e-12)(3.3071163e+02 -2.6010898e-13)
 (3.3071848e+02 3.2440114e-13)(3.3072532e+02 1.4270967e-12)(3.3073216e+02 -1.4266827e-12)
 (3.3073900e+02 -1.5561944e-12)(3.3074585e+02 -6.8105433e-13)(3.3075269e+02 -1.2299257e-12)
 (3.3075953e+02 -1.4596845e-12)(3.3076637e+02 1.4263701e-12)(3.3077322e+02 -1.2318583e-12)
 (3.3078006e+02 6.5483956e-14)(3.3078690e+02 -1.9449173e-12)(3.3079374e+02 2.5925247e-13)
 (3.3080059e+02 -1.2972568e-13)(3.3080743e+02 2.5972850e-13)(3.3081427e+02 1.4264372e-12)
 (3.3082111e+02 1.5567428e-12)(3.3082796e+02 1.7997501e-12)(3.3083480e+02 1.6217253e-12)
 (3.3084164e+02 -4.7072993e-13)(3.3084848e+02 1.1673815e-12)(3.3085533e+02 -9.4020652e-13)
 (3.3086217e+02 -9.8724261e-14)(3.3086901e+02 7.9138075e-12)(3.3087586e+02 5.4583015e-11)
 (3.3088270e+02 3.2323025e-10)(3.3088954e+02 1.7955032e-09)(3.3089638e+02 9.4431665e-09)
 (3.3090323e+02 4.7099195e-08)(3.3091007e+02 2.2278973e-07)(3.3091691e+02 9.9927010e-07)
 (3.3092375e+02 4.2497420e-06)(3.3093060e+02 1.7135386e-05)(3.3093744e+02 6.5499120e-05)
 (3.3094428e+02 2.3732336e-04)(3.3095112e+02 8.1499779e-04)(3.3095797e+02 2.6523087e-03)
 (3.3096481e+02 8.1785495e-03)(3.3097165e+02 2.3891126e-02)(3.3097849e+02 6.6102423e-02)
 (3.3098534e+02 1.7318783e-01)(3.3099218e+02 4.2955524e-01)(3.3099902e+02 1.0082878e+00)
 (3.3100587e+02 2.2389840e+00)(3.3101271e+02 4.7013654e+00)(3.3101955e+02 9.3297257e+00)
 (3.3102639e+02 1.7486328e+01)(3.3103324e+02 3.0928388e+01)(3.3104008e+02 5.1569847e+01)
 (3.3104692e+02 8.0953683e+01)(3.3105376e+02 1.1943151e+02)(3.3106061e+02 1.6520096e+02)
 (3.3106745e+02 2.1353703e+02)(3.3107429e+02 2.5667608e+02)(3.3108113e+02 2.8475390e+02)
 (3.3108798e+02 2.8789360e+02)(3.3109482e+02 2.5904373e+02)(3.3110166e+02 1.9668508e+02)
 (3.3110850e+02 1.0632927e+02)(3.3111535e+02 1.2382978e+02)(3.3112219e+02 -1.0630220e+02)
 (3.3112903e+02 -1.9665130e+02)(3.3113587e+02 -2.5899942e+02)(3.3114272e+02 -2.8783602e+02)
 (3.3114956e+02 -2.8468191e+02)(3.3115640e+02 -2.5659044e+02)(3.3116325e+02 -2.1344037e+02)
 (3.3117009e+02 -1.6509687e+02)(3.3117693e+02 -1.1932206e+02)(3.3118377e+02 -8.0833780e+01)
 (3.3119062e+02 -5.1414049e+01)(3.3119746e+02 -3.0660842e+01)(3.3120430e+02 -1.6914419e+01)

(3.3121114e+02 -8.0062842e+00)(3.3121799e+02 -1.6592221e+00)(3.3122483e+02 4.4779789e+00)
 (3.3123167e+02 1.3078563e+01)(3.3123851e+02 2.7524927e+01)(3.3124536e+02 5.2233649e+01)
 (3.3125220e+02 9.2650287e+01)(3.3125904e+02 1.5460255e+02)(3.3126588e+02 2.4275967e+02)
 (3.3127273e+02 3.5819468e+02)(3.3127957e+02 4.9550847e+02)(3.3128641e+02 6.4052913e+02)
 (3.3129326e+02 7.6996645e+02)(3.3130010e+02 8.5422698e+02)(3.3130694e+02 8.6367817e+02)
 (3.3131378e+02 7.7716290e+02)(3.3132063e+02 5.9011971e+02)(3.3132747e+02 3.1907883e+02)
 (3.3133431e+02 1.4339558e-01)(3.3134115e+02 -3.1880120e+02)(3.3134800e+02 -5.8986844e+02)
 (3.3135484e+02 -7.7695190e+02)(3.3136168e+02 -8.6351627e+02)(3.3136852e+02 -8.5411674e+02)
 (3.3137537e+02 -7.6990391e+02)(3.3138221e+02 -6.4050467e+02)(3.3138905e+02 -4.9550828e+02)
 (3.3139589e+02 -3.5820181e+02)(3.3140274e+02 -2.4275306e+02)(3.3140958e+02 -1.5455046e+02)
 (3.3141642e+02 -9.2491672e+01)(3.3142326e+02 -5.1834729e+01)(3.3143011e+02 -2.6587664e+01)
 (3.3143695e+02 -1.0963824e+01)(3.3144379e+02 1.0732073e-01)(3.3145064e+02 1.1163019e+01)
 (3.3145748e+02 2.6742409e+01)(3.3146432e+02 5.1921712e+01)(3.3147116e+02 9.2496986e+01)
 (3.3147801e+02 1.5447271e+02)(3.3148485e+02 2.4260505e+02)(3.3149169e+02 3.5801009e+02)
 (3.3149853e+02 4.9531008e+02)(3.3150538e+02 6.4034292e+02)(3.3151222e+02 7.6982104e+02)
 (3.3151906e+02 8.5414865e+02)(3.3152590e+02 8.6368719e+02)(3.3153275e+02 7.7727059e+02)
 (3.3153959e+02 5.9032570e+02)(3.3154643e+02 3.1936878e+02)(3.3155327e+02 4.8793162e-01)
 (3.3156012e+02 -3.1844510e+02)(3.3156696e+02 -5.8955258e+02)(3.3157380e+02 -7.7672866e+02)
 (3.3158065e+02 -8.6342834e+02)(3.3158749e+02 -8.5418768e+02)(3.3159433e+02 -7.7013235e+02)
 (3.3160117e+02 -6.4086415e+02)(3.3160802e+02 -4.9595307e+02)(3.3161486e+02 -3.5867752e+02)
 (3.3162170e+02 -2.4321035e+02)(3.3162854e+02 -1.5496214e+02)(3.3163539e+02 -9.2877941e+01)
 (3.3164223e+02 -5.2309445e+01)(3.3164907e+02 -2.7449879e+01)(3.3165591e+02 -1.2873125e+01)
 (3.3166276e+02 -4.1794036e+00)(3.3166960e+02 2.0012548e+00)(3.3167644e+02 8.3369934e+00)
 (3.3168328e+02 1.7185801e+01)(3.3169013e+02 3.0850786e+01)(3.3169697e+02 5.1559456e+01)
 (3.3170381e+02 8.1094776e+01)(3.3171065e+02 1.2011470e+02)(3.3171750e+02 1.6736463e+02)
 (3.3172434e+02 2.1917801e+02)(3.3173118e+02 2.6977365e+02)(3.3173803e+02 3.1275929e+02)
 (3.3174487e+02 3.4384932e+02)(3.3175171e+02 3.6414540e+02)(3.3175855e+02 3.8264623e+02)
 (3.3176540e+02 4.1631807e+02)(3.3177224e+02 4.8642458e+02)(3.3177908e+02 6.1101188e+02)
 (3.3178592e+02 7.9521185e+02)(3.3179277e+02 1.0227090e+03)(3.3179961e+02 1.2524512e+03)
 (3.3180645e+02 1.4237695e+03)(3.3181329e+02 1.4703971e+03)(3.3182014e+02 1.3401948e+03)
 (3.3182698e+02 1.0142184e+03)(3.3183382e+02 5.1779203e+02)(3.3184066e+02 -8.1594253e+01)
 (3.3184751e+02 -6.8997986e+02)(3.3185435e+02 -1.2112608e+03)(3.3186119e+02 -1.5717109e+03)
 (3.3186804e+02 -1.7363808e+03)(3.3187488e+02 -1.7126620e+03)(3.3188172e+02 -1.5415310e+03)
 (3.3188856e+02 -1.2813385e+03)(3.3189541e+02 -9.9070136e+02)(3.3190225e+02 -7.1585944e+02)
 (3.3190909e+02 -4.8493396e+02)(3.3191593e+02 -3.0851194e+02)(3.3192278e+02 -1.8411750e+02)
 (3.3192962e+02 -1.0175102e+02)(3.3193646e+02 -4.8375597e+01)(3.3194330e+02 -1.0314731e+01)
 (3.3195015e+02 2.6541137e+01)(3.3195699e+02 7.8225470e+01)(3.3196383e+02 1.6502143e+02)

(3.3197067e+02 3.1341452e+02)(3.3197752e+02 5.5606266e+02)(3.3198436e+02 9.2791536e+02)
 (3.3199120e+02 1.4569699e+03)(3.3199804e+02 2.1496465e+03)(3.3200489e+02 2.9735350e+03)
 (3.3201173e+02 3.8435953e+03)(3.3201857e+02 4.6200880e+03)(3.3202542e+02 5.1254665e+03)
 (3.3203226e+02 5.1819601e+03)(3.3203910e+02 4.6626570e+03)(3.3204594e+02 3.5402178e+03)
 (3.3205279e+02 1.9138468e+03)(3.3205963e+02 1.7437302e+01)(3.3206647e+02 -1.9134869e+03)
 (3.3207331e+02 -3.5398268e+03)(3.3208016e+02 -4.6622213e+03)(3.3208700e+02 -5.1814749e+03)
 (3.3209384e+02 -5.1249346e+03)(3.3210068e+02 -4.6195186e+03)(3.3210753e+02 -3.8430016e+03)
 (3.3211437e+02 -2.9729327e+03)(3.3212121e+02 -2.1490506e+03)(3.3212805e+02 -1.4563892e+03)
 (3.3213490e+02 -9.2733875e+02)(3.3214174e+02 -5.5542481e+02)(3.3214858e+02 -3.1251458e+02)
 (3.3215543e+02 -1.6333241e+02)(3.3216227e+02 -7.4474640e+01)(3.3216911e+02 -1.7855064e+01)
 (3.3217595e+02 3.0003787e+01)(3.3218280e+02 9.1018110e+01)(3.3218964e+02 1.8921974e+02)
 (3.3219648e+02 3.5344976e+02)(3.3220332e+02 6.1736995e+02)(3.3221017e+02 1.0150668e+03)
 (3.3221701e+02 1.5710945e+03)(3.3222385e+02 2.2854880e+03)(3.3223069e+02 3.1171120e+03)
 (3.3223754e+02 3.9718234e+03)(3.3224438e+02 4.7035654e+03)(3.3225122e+02 5.1347110e+03)
 (3.3225806e+02 5.0958998e+03)(3.3226491e+02 4.4767322e+03)(3.3227175e+02 3.2708667e+03)
 (3.3227859e+02 1.5969869e+03)(3.3228543e+02 -3.1657195e+02)(3.3229228e+02 -2.1825402e+03)
 (3.3229912e+02 -3.7253842e+03)(3.3230596e+02 -4.7479941e+03)(3.3231281e+02 -5.1721605e+03)
 (3.3231965e+02 -5.0416942e+03)(3.3232649e+02 -4.4918575e+03)(3.3233333e+02 -3.7002825e+03)
 (3.3234018e+02 -2.8381573e+03)(3.3234702e+02 -2.0360977e+03)(3.3235386e+02 -1.3704081e+03)
 (3.3236070e+02 -8.6709088e+02)(3.3236755e+02 -5.1620465e+02)(3.3237439e+02 -2.8866866e+02)
 (3.3238123e+02 -1.4998528e+02)(3.3238807e+02 -6.8588771e+01)(3.3239492e+02 -1.9082284e+01)
 (3.3240176e+02 1.8068847e+01)(3.3240860e+02 5.8563323e+01)(3.3241544e+02 1.1646575e+02)
 (3.3242229e+02 2.0543410e+02)(3.3242913e+02 3.3818062e+02)(3.3243597e+02 5.2379743e+02)
 (3.3244282e+02 7.6325601e+02)(3.3244966e+02 1.0443785e+03)(3.3245650e+02 1.3386200e+03)
 (3.3246334e+02 1.6025072e+03)(3.3247019e+02 1.7858530e+03)(3.3247703e+02 1.8465867e+03)
 (3.3248387e+02 1.7686553e+03)(3.3249071e+02 1.5763616e+03)(3.3249756e+02 1.3374934e+03)
 (3.3250440e+02 1.1499991e+03)(3.3251124e+02 1.1126500e+03)(3.3251808e+02 1.2872626e+03)
 (3.3252493e+02 1.6656177e+03)(3.3253177e+02 2.1552974e+03)(3.3253861e+02 2.5938610e+03)
 (3.3254545e+02 2.7910772e+03)(3.3255230e+02 2.5876837e+03)(3.3255914e+02 1.9110073e+03)
 (3.3256598e+02 8.0678340e+02)(3.3257283e+02 -5.6596212e+02)(3.3257967e+02 -1.9765311e+03)
 (3.3258651e+02 -3.1862390e+03)(3.3259335e+02 -4.0115995e+03)(3.3260020e+02 -4.3657983e+03)
 (3.3260704e+02 -4.2671496e+03)(3.3261388e+02 -3.8166234e+03)(3.3262072e+02 -3.1571939e+03)
 (3.3262757e+02 -2.4315261e+03)(3.3263441e+02 -1.7511603e+03)(3.3264125e+02 -1.1829061e+03)
 (3.3264809e+02 -7.5073937e+02)(3.3265494e+02 -4.4703590e+02)(3.3266178e+02 -2.4618153e+02)
 (3.3266862e+02 -1.1541410e+02)(3.3267546e+02 -2.0441539e+01)(3.3268231e+02 7.4331302e+01)
 (3.3268915e+02 2.0967529e+02)(3.3269599e+02 4.3729526e+02)(3.3270283e+02 8.2438578e+02)
 (3.3270968e+02 1.4529178e+03)(3.3271652e+02 2.4090468e+03)(3.3272336e+02 3.7591133e+03)

(3.3273021e+02 5.5126535e+03)(3.3273705e+02 7.5798533e+03)(3.3274389e+02 9.7390142e+03)
 (3.3275073e+02 1.1634541e+04)(3.3275758e+02 1.2822612e+04)(3.3276442e+02 1.2867367e+04)
 (3.3277126e+02 1.1468201e+04)(3.3277810e+02 8.5779060e+03)(3.3278495e+02 4.4640799e+03)
 (3.3279179e+02 -3.1993742e+02)(3.3279863e+02 -5.0559035e+03)(3.3280547e+02 -9.0373744e+03)
 (3.3281232e+02 -1.1742416e+04)(3.3281916e+02 -1.2944183e+04)(3.3282600e+02 -1.2727688e+04)
 (3.3283284e+02 -1.1418743e+04)(3.3283969e+02 -9.4617732e+03)(3.3284653e+02 -7.2943481e+03)
 (3.3285337e+02 -5.2566586e+03)(3.3286022e+02 -3.5524142e+03)(3.3286706e+02 -2.2558963e+03)
 (3.3287390e+02 -1.3468585e+03)(3.3288074e+02 -7.5284470e+02)(3.3288759e+02 -3.8411085e+02)
 (3.3289443e+02 -1.5451909e+02)(3.3290127e+02 1.1488210e+01)(3.3290811e+02 1.8198046e+02)
 (3.3291496e+02 4.2556540e+02)(3.3292180e+02 8.1906550e+02)(3.3292864e+02 1.4495688e+03)
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 (3.3301760e+02 -5.0511190e+03)(3.3302444e+02 -9.0333471e+03)(3.3303128e+02 -1.1739708e+04)
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(3.3587097e+02 3.5820181e+02)(3.3587781e+02 4.9550828e+02)(3.3588465e+02 6.4050467e+02)
(3.3589150e+02 7.6990391e+02)(3.3589834e+02 8.5411674e+02)(3.3590518e+02 8.6351627e+02)
(3.3591202e+02 7.7695190e+02)(3.3591887e+02 5.8986844e+02)(3.3592571e+02 3.1880120e+02)
(3.3593255e+02 -1.4339558e-01)(3.3593939e+02 -3.1907883e+02)(3.3594624e+02 -5.9011971e+02)
(3.3595308e+02 -7.7716290e+02)(3.3595992e+02 -8.6367817e+02)(3.3596676e+02 -8.5422698e+02)
(3.3597361e+02 -7.6996645e+02)(3.3598045e+02 -6.4052913e+02)(3.3598729e+02 -4.9550847e+02)
(3.3599413e+02 -3.5819468e+02)(3.3600098e+02 -2.4275967e+02)(3.3600782e+02 -1.5460255e+02)
(3.3601466e+02 -9.2650287e+01)(3.3602151e+02 -5.2233649e+01)(3.3602835e+02 -2.7524927e+01)
(3.3603519e+02 -1.3078563e+01)(3.3604203e+02 -4.4779789e+00)(3.3604888e+02 1.6592221e+00)
(3.3605572e+02 8.0062842e+00)(3.3606256e+02 1.6914419e+01)(3.3606940e+02 3.0660842e+01)
(3.3607625e+02 5.1414049e+01)(3.3608309e+02 8.0833780e+01)(3.3608993e+02 1.1932206e+02)
(3.3609677e+02 1.6509687e+02)(3.3610362e+02 2.1344037e+02)(3.3611046e+02 2.5659044e+02)
(3.3611730e+02 2.8468191e+02)(3.3612414e+02 2.8783602e+02)(3.3613099e+02 2.5899942e+02)
(3.3613783e+02 1.9665130e+02)(3.3614467e+02 1.0630220e+02)(3.3615152e+02 -1.2382978e-02)
(3.3615836e+02 -1.0632927e+02)(3.3616520e+02 -1.9668508e+02)(3.3617204e+02 -2.5904373e+02)
(3.3617889e+02 -2.8789360e+02)(3.3618573e+02 -2.8475390e+02)(3.3619257e+02 -2.5667608e+02)
(3.3619941e+02 -2.1353703e+02)(3.3620626e+02 -1.6520096e+02)(3.3621310e+02 -1.1943151e+02)
(3.3621994e+02 -8.0953683e+01)(3.3622678e+02 -5.1569847e+01)(3.3623363e+02 -3.0928388e+01)
(3.3624047e+02 -1.7486328e+01)(3.3624731e+02 -9.3297257e+00)(3.3625415e+02 -4.7013654e+00)
(3.3626100e+02 -2.2389840e+00)(3.3626784e+02 -1.0082878e+00)(3.3627468e+02 -4.2955524e-01)
(3.3628152e+02 -1.7318783e-01)(3.3628837e+02 -6.6102423e-02)(3.3629521e+02 -2.3891126e-02)
(3.3630205e+02 -8.1785495e-03)(3.3630890e+02 -2.6523087e-03)(3.3631574e+02 -8.1499779e-04)
(3.3632258e+02 -2.3732336e-04)(3.3632942e+02 -6.5499121e-05)(3.3633627e+02 -1.7135387e-05)
(3.3634311e+02 -4.2497408e-06)(3.3634995e+02 -9.9927115e-07)(3.3635679e+02 -2.2278817e-07)
(3.3636364e+02 -4.7098315e-08)(3.3637048e+02 -9.4415535e-09)(3.3637732e+02 -1.7938208e-09)
(3.3638416e+02 -3.2209422e-10)(3.3639101e+02 -5.5164244e-11)(3.3639785e+02 -8.2201971e-12)
(3.3640469e+02 -3.2639357e-12)(3.3641153e+02 6.1870745e-13)(3.3641838e+02 -8.0704766e-13)
(3.3642522e+02 7.6695983e-13)(3.3643206e+02 6.7549162e-13)(3.3643891e+02 -6.8957416e-13)
(3.3644575e+02 1.7363205e-13)(3.3645259e+02 -3.5202526e-13)(3.3645943e+02 -8.3876031e-13)
(3.3646628e+02 -4.1900819e-13)(3.3647312e+02 -1.2692257e-12)(3.3647996e+02 -6.6484576e-14)
(3.3648680e+02 1.1496180e-12)(3.3649365e+02 2.5987277e-13)(3.3650049e+02 -5.8597653e-13)
(3.3650733e+02 -4.2842005e-13)(3.3651417e+02 -1.7869303e-15)(3.3652102e+02 2.4720951e-13)

(3.3652786e+02 -1.1407693e-12)(3.3653470e+02 -2.3329469e-12)(3.3654154e+02 -9.6818043e-13)
 (3.3654839e+02 -1.6508377e-12)(3.3655523e+02 -6.4338599e-13)(3.3656207e+02 -2.4002259e-12)
 (3.3656891e+02 -1.8664843e-12)(3.3657576e+02 -9.0200111e-13)(3.3658260e+02 -2.9592948e-13)
 (3.3658944e+02 -1.3711815e-12)(3.3659629e+02 1.1526402e-12)(3.3660313e+02 1.4363380e-12)
 (3.3660997e+02 4.7009632e-13)(3.3661681e+02 5.5037074e-14)(3.3662366e+02 -6.0592187e-13)
 (3.3663050e+02 1.2709548e-12)(3.3663734e+02 4.7917410e-13)(3.3664418e+02 2.0036194e-12)
 (3.3665103e+02 2.1751686e-12)(3.3665787e+02 3.1651895e-12)(3.3666471e+02 1.9829845e-12)
 (3.3667155e+02 5.4003196e-13)(3.3667840e+02 -2.9416167e-13)(3.3668524e+02 6.9353442e-14)
 (3.3669208e+02 -9.2052878e-13)(3.3669892e+02 -4.3256970e-14)(3.3670577e+02 -1.2237247e-12)
 (3.3671261e+02 -3.6754287e-13)(3.3671945e+02 -6.8813741e-13)(3.3672630e+02 -1.6914921e-12)
 (3.3673314e+02 -2.5374196e-12)(3.3673998e+02 -1.3266150e-13)(3.3674682e+02 -1.9866759e-12)
 (3.3675367e+02 6.3182103e-13)(3.3676051e+02 -7.2095094e-13)(3.3676735e+02 1.4813363e-13)
 (3.3677419e+02 3.8049638e-13)(3.3678104e+02 -1.3397735e-12)(3.3678788e+02 3.4689067e-13)
 (3.3679472e+02 -1.8675183e-12)(3.3680156e+02 -1.0678405e-12)(3.3680841e+02 1.2884404e-12)
 (3.3681525e+02 -1.8178286e-12)(3.3682209e+02 -1.1528186e-12)(3.3682893e+02 -2.1764823e-12)
 (3.3683578e+02 -5.8444646e-13)(3.3684262e+02 -2.5511424e-12)(3.3684946e+02 -1.0664071e-12)
 (3.3685630e+02 -4.7130912e-14)(3.3686315e+02 1.8148028e-12)(3.3686999e+02 1.9425003e-12)
 (3.3687683e+02 2.5533923e-12)(3.3688368e+02 7.6064634e-13)(3.3689052e+02 1.7720063e-12)
 (3.3689736e+02 1.1512942e-12)(3.3690420e+02 2.7750209e-12)(3.3691105e+02 1.0296889e-12)
 (3.3691789e+02 1.1159626e-12)(3.3692473e+02 9.2557218e-13)(3.3693157e+02 7.0066539e-13)
 (3.3693842e+02 -2.2658625e-12)(3.3694526e+02 -1.4247302e-12)(3.3695210e+02 -1.2583896e-12)
 (3.3695894e+02 -4.3885618e-13)(3.3696579e+02 -2.4968984e-12)(3.3697263e+02 -2.9041191e-12)
 (3.3697947e+02 1.0812935e-12)(3.3698631e+02 -2.9382381e-13)(3.3699316e+02 -2.9367226e-13)
 (3.3700000e+02 -5.6053008e-13)(700000e+02 -5.6053008e-1)

Please carefully analyze the spectrum and determine the value of y .

A. 5

B. 7

C. 9

D. 11

E. 13

F. 15

G. 17

H. 19

I. 21

J. 23

K. 25

L. 27

M. All other answers are incorrect.

N. 6

O. 8

P. 10

Q. 12

R. 14

S. 16

Answer

Correct Answer: **C**

Detailed Explanation

Analysis of the spectrum reveals that the EPR spectrum is overall split into a septet (with an integral ratio of 1:6:15:20:15:6:1).

CHECKPOINT

2 PTS

Overall split into a septet (with an integral ratio of 1:6:15:20:15:6:1)

Each peak is further split into a quartet (with an integral ratio of 1:3:3:1).

CHECKPOINT

2 PTS

Each peak is further split into a quartet (with an integral ratio of 1:3:3:1)

Therefore, the free radical undergoes a first splitting by 6 equivalent protons and a second splitting by 3 equivalent protons, for a total of 9 protons, hence $y = 9$.

CHECKPOINT

1 PTS

Total of 9 protons

